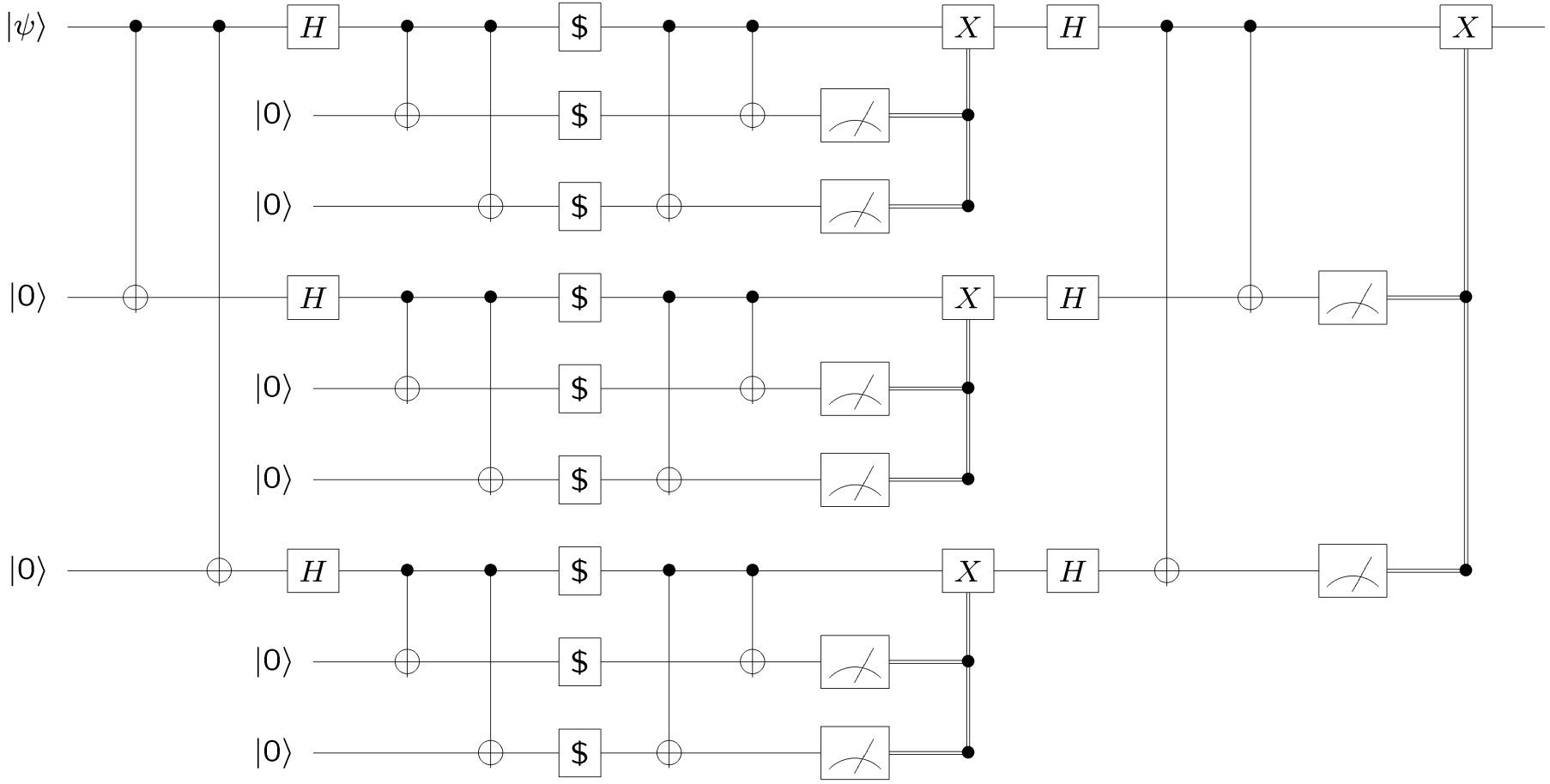
# Designing Quantum Computers

## Dave Bacon



# The Quantum Dream




Quantum computer:

a computer whose physical operation exploits the quantum laws of physics

Quantum machines will change our lives.

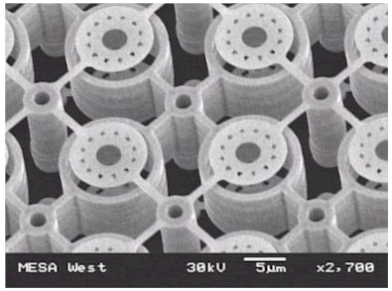
- Break public key cryptosystems
- Simulate quantum systems
  revolutionize chemistry, biology, material science
- ??? [This is what I spend about half my time working on]

# The Quest

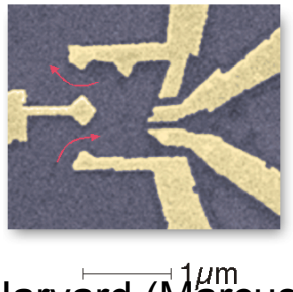
Building a quantum computer:

100s of physicists (worldwide) working on hardware for quantum computers.

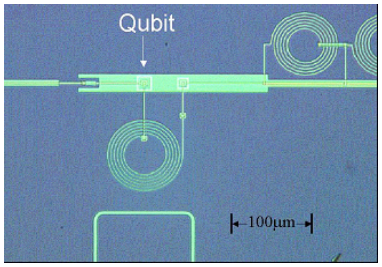
Cusp of large efforts to build scalable quantum computers.



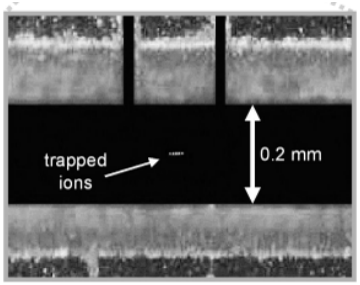

Sandia National lab
ion traps




Harvard (Marcus)
quantum dots




UCSB (Martinis)
Josphenson junctions




NIST (Wineland)
Ion traps

University of Washington
Computer Science & Engineering



Oskin

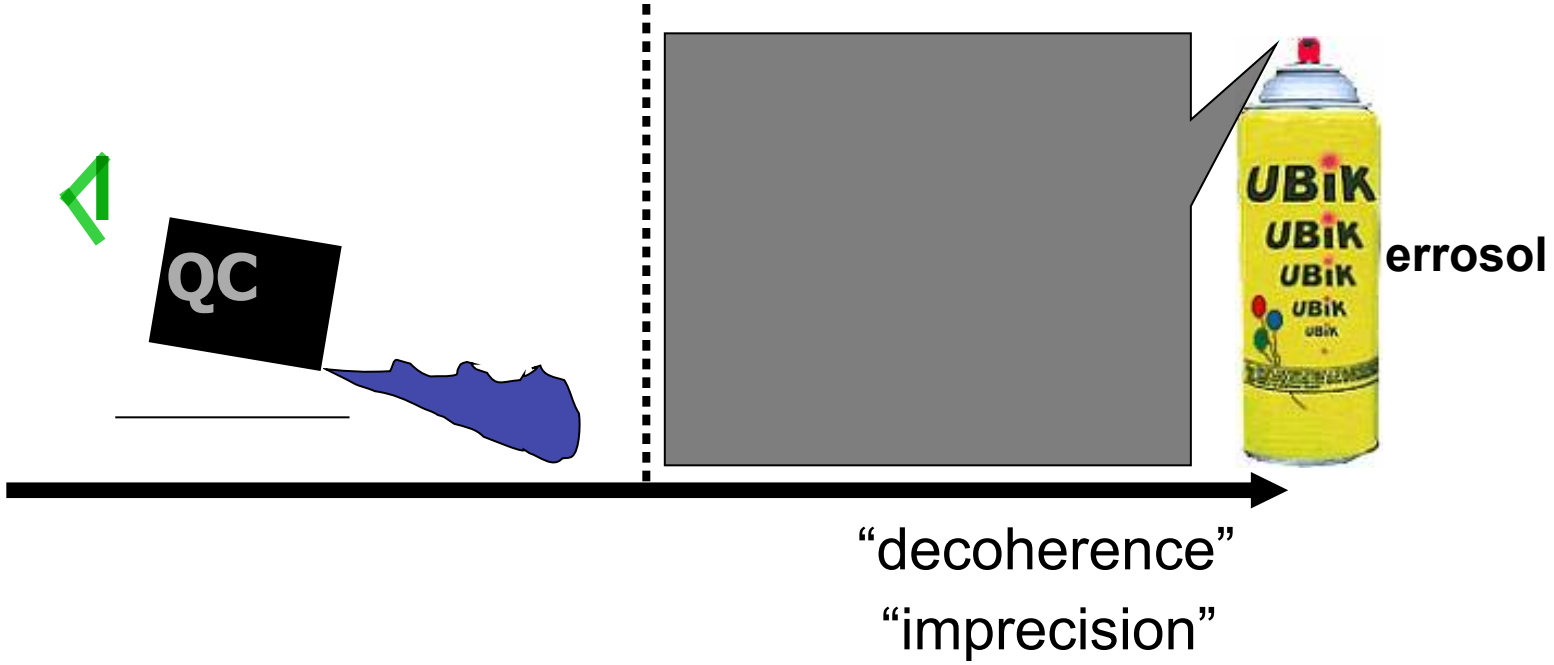


Bacon

One of only a very few groups who are poised to meaningfully contribute to designing this scaleup:

**Quantum Computer Architectures**

# A Quantum Theorem




**Threshold Theorems for Fault-Tolerant Quantum Computation**

[a partial list from the mid 90s: Aharonov, Ben-Or, Kitaev, Knill, LaFlamme, Zurek, Shor, DiVincenzo, Gottesman, Preskill]

If decoherence is weak enough and control is precise enough, then a quantum computation to a desired accuracy can be carried out with an overhead polynomial in the log of the accuracy.

# A Theorem Does Not a Computer Make

What I do: a portfolio of two

1. Architecture friendly quantum error correcting codes

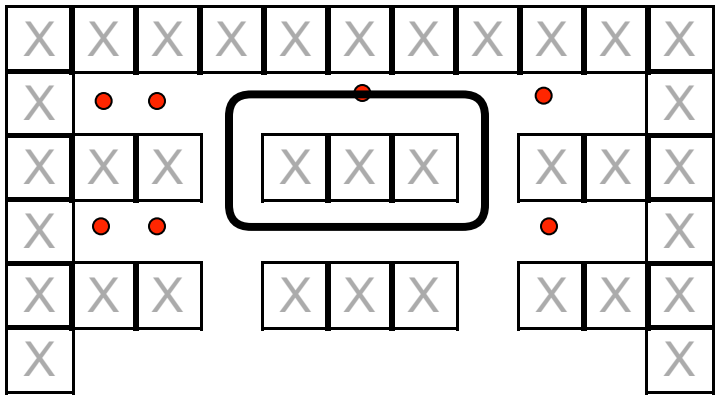
quantum error correcting codes

hardware constraints
hardware specifications

spatial locality

THEORY

ARCHITECTURE



XXX
XXX
| | |

measurement

# A Theorem Does Not a Computer Make

What I do: a portfolio of two

2. Self-correcting quantum micro-architectures

**Classical computers are fault-tolerant due to the physics of the device. The physics enacts error correction.**

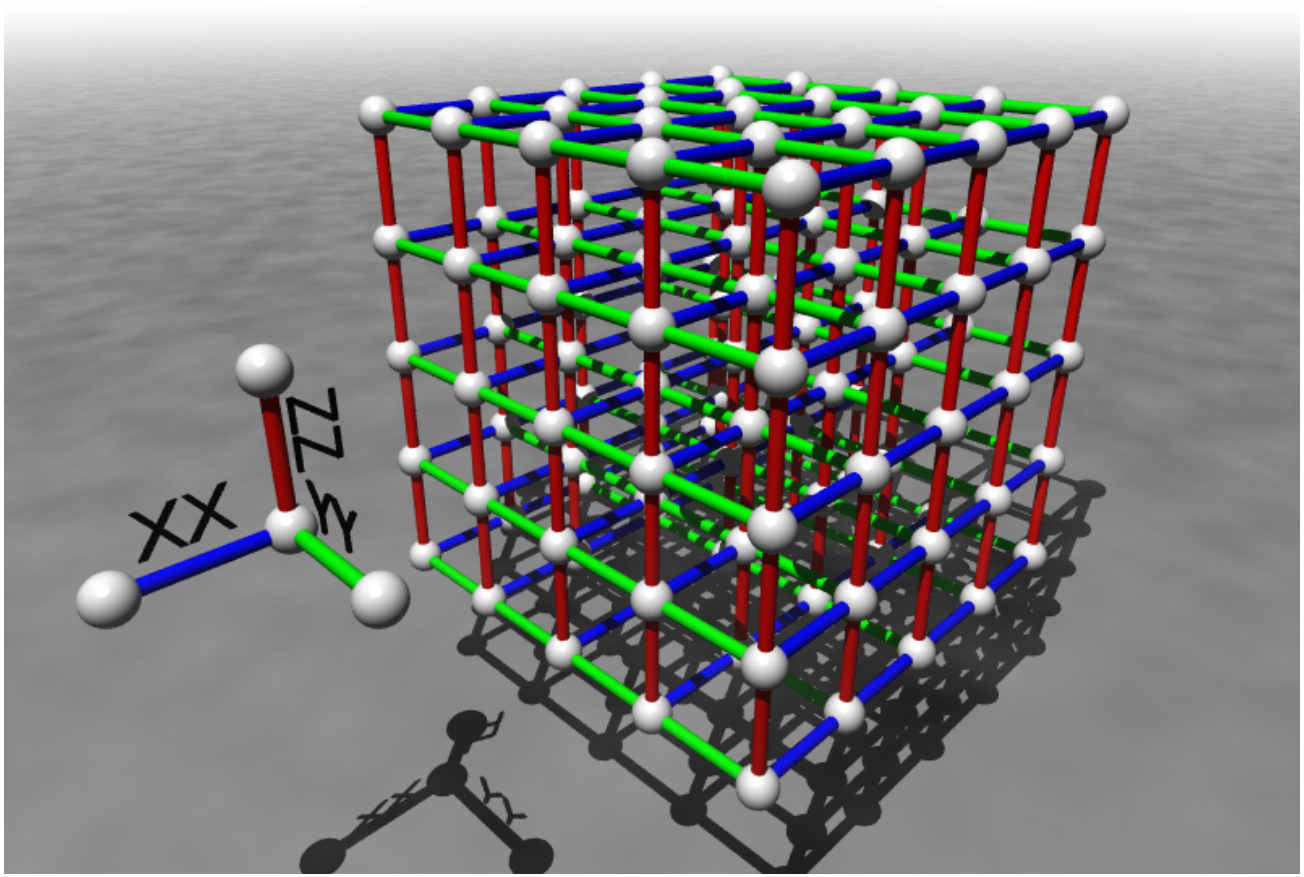
Hard Drive 

Integrated Circuit 

**Can we engineer physical systems whose physics naturally enacts quantum error correction?**

# Toward Quantum Hard Drives

Self Correcting Quantum Memories



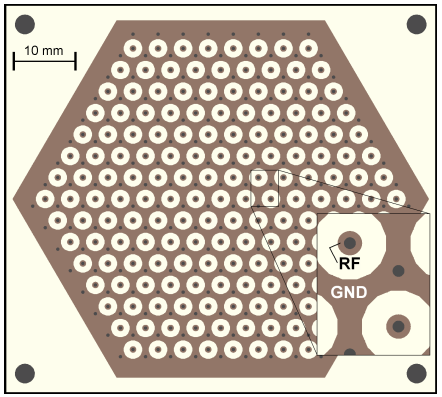
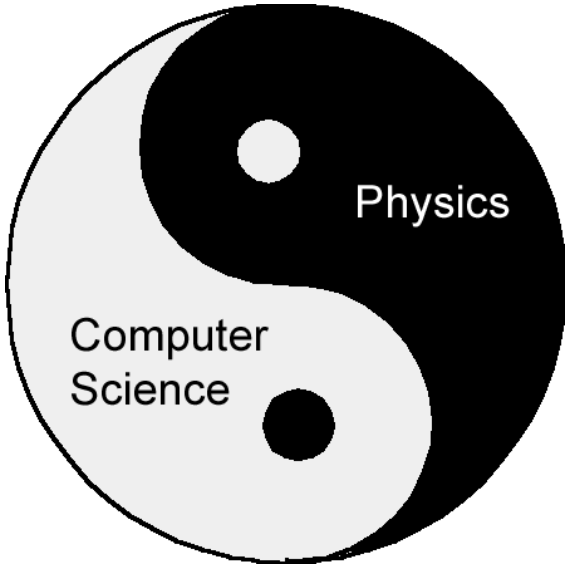




**Figure 2. Hexagonal planar ion trap. 2D mult-zone ion traps allow one to trap ions in any spatial arrangement including a disordered lattice.**

(ion trap picture courtesy Ken Brown, MIT)

# Quantum Computers








…an invitation to help found a new field

# One Reason To Build

188198812920607963838697239461650439807163563379417382700763356422988859715234665485319060606504743045317388011303396716199623212057340318795506569622130516875930765025705 9

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472772146107435302536223071973048224632914695302097116459852171130520711256363590397527



Peter Shor

Shor's algorithm (1994) breaks the most widely used public key cryptosystems (RSA, Diffie-Hellman)